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The Effects of Social Skill Interventions for Adolescents With Autism: A Meta-Analysis

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Abstract

Communicating with peers often poses challenges for individuals with autism spectrum disorder (ASD); these social interactions, however, can be especially difficult during the teenage years. For many adolescents with ASD, peer interaction is often limited or nonexistent, even for those taught in general education settings. The purpose of this meta-analysis was to investigate social interaction interventions for adolescents with ASD within public school settings, with a focus on studies that utilized single-case design. Studies were evaluated per the What Works Clearinghouse guidelines. This meta-analysis includes 24 studies with participants with ASD ranging in ages from 13 to 21 years. The effects of social skill interventions were evaluated using visual analysis, Tau-*U*, and a parametric effect size (*d*-Hedges–Pustejovsky–Shadish [DHPS]). Combined effect sizes were calculated and compared. Results provide evidence that interventions can produce positive change in social behavior of adolescents with ASD. Effect sizes were generally large. Interpretations and implications are discussed.

Keywords

autism spectrum disorder, social skills, social interactions, adolescents

Positive social interactions with peers are critical to the quality of life for all individuals, and are especially important during the adolescent years (Rubin et al., 2011). These interactions are an important part of the developmental process and impact not only success and well-being during the school years, but in adult life as well (Wentzel et al., 2012). It is through interactions with peers that important communication skills are learned and a personal identity is formed (Rubin et al., 2011). Relationships with peers support the development of cognitive, emotional, and social competence (Bagwell & Schmidt, 2011).

During adolescence, there are increased demands for communication as social networks are formed around mutual interests (Bagwell & Schmidt, 2011; Wentzel et al., 2012). Conversations typically occur at a fast rate and within a group of peer communication partners, as teenagers create a better understanding of their life experiences by discussing them with others (Rubin et al., 2011). The nature of this communication is considerably different from the social exchanges observed with young children; interactions become complicated, and the understanding of one's self and others intensifies (Davis et al., 2010). Meaningful participation during adolescent interactions necessitates the ability to interpret and use nonverbal skills (e.g., body orientation and eye contact), respond contingently, and engage in threaded reciprocal conversation (Paul et al., 2009). Adolescence is also a time of change and challenge for peer relationships (Rubin et al., 2011), and those who struggle with peer relationships are more likely to report feelings of loneliness and depression in adult life (Wentzel et al., 2012).

The challenges of social interactions in the adolescent years are compounded for adolescents with autism spectrum disorder (ASD). Deficits in social skills are a defining characteristic of ASD (American Psychiatric Association, 2013), and many persons with ASD experience challenges in interpreting nonverbal cues, initiating interactions, responding to others, and maintaining a topic in an ongoing conversation (Paul et al., 2009). Social interaction may be particularly difficult for those adolescents with ASD and *complex communication needs*—that is, individuals for

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whom speech does not meet all of their communication needs (Beukelman & Mirenda, 2013). Although some individuals with ASD make use of speech to interact with others, approximately 40% of individuals with ASD do not develop functional or fluent speech (Centers for Disease Control and Prevention, n.d.). These individuals face additional challenges such as social isolation, difficulty creating and maintaining social networks, and limited opportunities to engage in typical adolescent vocational and recreational activities (Carter et al., 2014; Smith, 2015).

Social interactions are often limited between the students with disabilities and their typically developing peers (Feldman et al., 2015). According to the U.S. Department of Education (2011), approximately 60% of adolescents with ASD spend 40% or more of their school day in the general education setting in public schools. However, the meaningfulness of these inclusion opportunities remains unclear. For example, Chung et al. (2012) observed 16 learners with severe disabilities in general education classrooms for more than 50 hr and reported that two of the four adolescents with ASD and complex communication needs never interacted with peers, even when they were in close physical proximity. Chung et al. concluded that unless peer interactions were specifically planned for and supported by educational staff, students with ASD and complex communication needs interacted with peers infrequently, if at all.

Social skill interventions for young children with ASD have received substantial research attention. Recent systematic reviews suggest that interventions that address the behavior of both the child with ASD and their typically developing peer can produce positive outcomes (Chapin et al., 2018; Hansen et al., 2014). Recently, Gates et al. (2017) conducted a meta-analysis of group-based social skill interventions for children and youth with ASD and reported that the interventions were moderately effective. Ke et al. (2018) also reported positive findings in their review of social skills interventions for youth and adults with ASD; however, they found mixed results in regards to generalization of social skills.

Although both of these reviews provide evidence of the benefits of social skill intervention, there are important questions related to social skills interventions for adolescents with ASD in public school settings that have not been addressed. For example, both Gates et al. (2017) and Ke et al. (2018) included studies occurring outside of a typical school setting (e.g., clinic). Gates et al. (2017) excluded single-case design studies, a frequently used approach in applied settings for individuals with low-incidence disabilities (Gast et al., 2018; What Works Clearinghouse, 2014). Ke et al. (2018) excluded participants with moderate or severe intellectual disability (ID), which is a commonly occurring comorbidity (Centers for Disease Control, 2012; Matson & Shoemaker, 2009).

Purpose of the Current Study

In this meta-analysis, we reviewed single-case research studies of social skill interventions for adolescents with ASD in public school settings. We sought to answer the following four research questions:

Research Question 1 (RQ1): What social skill interventions have been studied for adolescents with ASD in schools?
Research Question 2 (RQ2): What specific social skills (i.e., outcome variables) have been targeted?
Research Question 3 (RQ3): What is the quality of the designs and strength of evidence for each study according to the What Works Clearinghouse (WWC) standards?
Research Question 4 (RQ4): What are the effects of studies as measured by two effect size estimates?

Method

The first step in identifying the evidence base of social interaction interventions for adolescents with ASD in school settings was a systematic search of all published literature through 2018. The final search was conducted on December 31, 2018. We searched three databases: PsychINFO via ProQuest Education Journal, PubMed, and Education Resource Information Center (ERIC) using a combination of the following key terms: ASD, social interactions, and middle/high school. For each database, a Boolean phrase (e.g., ASD AND social interaction* AND high school) was used that varied slightly depending on specific requirements of the database search engine. The search terms were developed with an expert librarian from the University. The Boolean phrase was searched in the title and abstract fields. Only English-language peer-reviewed studies were included in the search. Ancestral searches were conducted by screening the reference lists of identified studies. See Supplemental Materials for a detailed description of the search and a graphic representation of articles that were screened.

Procedure

Title/abstract review. Titles and abstracts of each article were reviewed. Articles were excluded if they (a) did not include at least one participant with ASD, (b) did not contain at least one dependent variable related to social interactions, (c) did not contain at least one independent variable focused on an intervention for participants with disabilities, or (d) did not utilize a single-case design. Peer-reviewed articles, articles printed in English and articles published prior to 2018 were eligible for inclusion. If it was not possible to determine whether the exclusion criteria were met from the information provided in the title and abstract, the article was reviewed at the full-text level.

Full-text review. We reviewed articles that were not excluded at the title/abstract level in greater depth at the full-text level. To be included the study had to (a) include participant(s) with a medical diagnosis or special education eligibility of ASD between the ages of 13 and 21 years, (b) measure at least one dependent variable related to social interaction outcomes, (c) be conducted in a public school during typical school hours, (d) utilize a single-case design, and (e) be published in a peer reviewed journal printed in English. Social interactions were defined by Walton and Ingersoll (2013) as social behaviors, " . . . that actively support interactions with peers or adults, including conversation, imitation, social initiations, and responses, interest in other people, socially directed communication, and nonverbal social behaviors (e.g., eye contact, proximity to others)" (p. 596). Studies that examined social skill components that did not match our definition were not included (e.g., social attributions; Bhatara et al., 2009). A total of 26 articles met the criteria at this stage.

Design quality. All articles that met the full-text review inclusion criteria were evaluated for design quality using the basic design standards for single-case research by WWC (Kratochwill et al., 2010). Studies were evaluated based on intervention implementation, inter-assessor agreement, and three opportunities to demonstrate an effect on the dependent variable over time. We visually analyzed the data for changes in level, trend, variability, immediacy of an effect, overlap, and consistency between similar phases. We applied the recommended steps for visual analysis (observation of a predictable baseline pattern, examination of data within each phase, comparison of phases to determine whether an effect was present, and confirmation of the demonstrated effect across at least 3 times). Studies with graphs that met the WWC visual analysis requirements and included interobserver agreement (IOA) at more than 20% met evidence standards. Studies in which the graph met none of the WWC visual analysis requirements, the IOA was less than 20%, or the intervention was not systematically manipulated, did not meet evidence standards. A study met evidence standards with reservations if the graph met some but not all of the WWC requirements and IOA was more than 20%. Next, each study was coded as meets standards without reservations, meets standards with reservations, or does not meet standards. Of the 26 included studies, eight studies met the WWC design standards (31%), 16 met with reservations (54%), and two did not meet the standards (15%). At this stage, two studies were eliminated because they did not meet standards. Twenty-four articles met the design quality criteria and were included in this review. In addition, following the WWC model, visual analysis was conducted for the remaining 24 studies that met standards *with* and *without* reservations. See Supplemental Materials for a graphic representation.

Variable Codes

Studies were coded on eight primary variables: (a) participant characteristics, (b) mode of communication, (c) setting, (d) type of social interaction intervention, (e) dependent variables, (f) experimental design, (g) design quality, and (h) strength of evidence.

Participant demographics. We coded the age, gender, and race/ethnicity of participants. Next, diagnosis status was coded based on the primary disability (e.g., ASD, ASD and co-morbid diagnosis, or high-functioning autism) status of the participant. Communication modality was coded for each participant as (a) having the ability to use speech during interactions with others without additional supports or (b) having complex communication needs or exhibiting severe limitations in speech (e.g., two- to four-word sentences, single words or short phrases, or echolalia). In addition, based on author description, the setting where the intervention took place was coded as special education classroom, general education classroom, cafeteria, common spaces (e.g., hallway and courtyard), or multiple settings.

Independent variables. Based on author description, we coded intervention strategies used to teach social skills. Three intervention categories were identified: (a) peer-directed interventions (i.e., studies in which the primary instruction was provided to the peers, not the participants); (b) learner with ASD-directed interventions (i.e., the intervention was primarily facilitated by an adult and instruction was provided to the participant with ASD); and (c) a combined-approach intervention (i.e., instruction was provided for both peers and the participants with ASD).

In *peer-directed* interventions, typically developing peers were the initial focus of the intervention, and they were trained or encouraged to model and prompt desired social skills for students with ASD (e.g., Chan et al., 2009). Studies were coded as peer-directed interventions if an adult, typically a teacher or researcher provided either (a) organized instruction (i.e., direct teaching or instruction to peer partners on social interactions) or (b) social supports (i.e., minimal social assistance to the peer to promote interaction, such as directing the peer to sit in proximity to a learner with ASD, and/or cueing the peer to interact with a focal student). Studies were coded as *learner with ASD-directed* interventions when an adult either provided (a) direct teaching or instruction to the participants with ASD or (b) social assistance to the learner with ASD to promote interaction (e.g., assisted the learner with ASD in joining a social group, provided limited assistance in running of social group). *Combined-approach* interventions included components of both peer-directed and learner with ASD-directed interventions.

Dependent variables. We coded studies for the following social-behavioral dependent variables: general communication acts, initiations, responses, commenting, others-focused conversation, follow-up questions, and engagement. The behavioral dimension of measurement used for the dependent variable was also coded (e.g., frequency of initiations and duration of conversation).

Design. We coded the type of single-case design conducted in each study (i.e., reversal/withdrawal), multiple baseline across participants (i.e., introduction of the intervention was staggered across participants), and multiple baseline across behaviors (i.e., introduction of the intervention was staggered across selected behaviors).

Reliability. We coded whether IOA was reported by the authors in each study and also mean IOA based on percentage (i.e., average of 90% or better and average of 80% or better). We also coded whether the authors for each study reported procedural integrity and the mean procedural fidelity score.

Social validity. We coded whether the authors collected and reported social validity data. Social validity data were further analyzed to assess who reported social validity (i.e., participants, peers, and staff members), the type of measure used to evaluate social validity (i.e., interviews and surveys), and finally, whether the social validity results were positive, negative, neutral, or mixed among those from whom it was assessed.

Strength of evidence. As indicated above in design quality, each phase in each graph was evaluated for a predictable pattern of data in baseline as well as changes in level, trend, and variability within each phase, immediacy of effect, and at least three demonstrations of an effect (What Works Clearinghouse, 2014). For studies that conducted multiple baseline across skills designs, we only evaluated graphs that included participants with ASD (e.g., Morrison et al., 2001). If the study made use of a multiple baseline across participants design, the whole graph was evaluated to conduct a vertical analysis (e.g., Hughes et al., 2011) to evaluate experimental control. Visual analysis results fell into one of the following three categories: strong evidence, moderate evidence, or no evidence. We adhered to the WWC guidelines to evaluate the strength of evidence. Strong evidence was defined as at least three demonstrations of an intervention effect with no noneffects. If the single-case design did not provide three demonstrations of an effect, then there was no evidence of a functional relation. If a study provided three demonstrations of an effect and also included at least one demonstration of a noneffect, there was moderate evidence of a functional relation.

Data Extraction

The first author used WebPlot Digitizer 2.0 to retrieve the raw single-case data from the included studies. WebPlot Digitizer is a data retrieval software program that is reliable and free of charge (Moeyaert et al., 2016). The data were extracted by importing the graph into the software program, defining a coordinate system, and manually selecting each data point. Two columns of values were formed. The first column, the x values, represent time and the second column, the y values, represent the dependent variable outcomes value. These data were then exported into a Microsoft Excel® file. A doctoral student who was blind to the study purpose retrieved at least 20% of the data. Interobserver reliability was 99% agreement for 22% of the graphs.

Data Analysis Procedures

Tau-U. WWC recommends calculation of effect size for studies that qualify as either strong or moderate evidence. However, for the purposes of the current meta-analysis, all studies that met methodological standards (with or without reservations) were included in the effect size calculation regardless of the strength of evidence. We modified the recommendation from the WWC guidelines to address issues related to inflation. Despite meta-analyses of single-case designs becoming more common, there is currently no consensus on the best effect size metric (Manolov & Moeyaert, 2017). Based on the lack of consensus, it is recommended that more than one effect size be used. Therefore, we calculated Tau-*U* as a nonparametric overlap index (Parker et al., 2011) and *d*-Hedges–Pustejovsky–Shadish (DHPS) effect size (Hedges et al., 2012) as a parametric effect size metric. We chose Tau-*U* because Tau-*U* compares all data points, considers baseline trend, and follows the sampling distribution, so that p values and confidence intervals can be estimated for the metric to be used in a meta-analysis. With the extracted data, all A–B contrasts relevant to social interactions were compared.

The nonparametric measure Tau-*U* was calculated using the online Tau-*U* calculator (Vannest et al., 2016). Tau-*U* is an effect size measure that calculates nonoverlapping data with baseline, has limited ability to control for baseline trend, and has effect sizes that range from 0 to 1. We did not need to control for baseline trend for any of the studies. Tau-*U* effect sizes are typically interpreted as small (.65 or lower), medium to large (.66–.92), or large (.93–1.0) (Vannest & Ninci, 2015). In instances where subsequent phases were implemented in a sequential order (e.g., providing training to the participant, and then providing training to the peer), data were treated as a packaged intervention, and thus combined for comparison to calculate Tau-*U* (e.g., Reilly et al., 2014). We made this decision because the B and C phases were similar (e.g., Training 1 and Training 2) and

predicted to produce the same effect on the behavior (i.e., an increase from baseline, not a reversal or secondary effect). The intervention was a sequential package where generalization was programmed. We were interested in determining the overall effects of the intervention package. Combining BC contrasts with AB contrasts may have resulted in an artificially low score as the high degree of predicted overlap between BC would decrease the weighted average when combined with AB contrast. In cases where both the researcher and a peer collected data (e.g., Morrison et al., 2001), only data collected by the researcher was examined. For each study, the weighted Tau-*U* was calculated for the entirety of the study. In addition, a weighted Tau-*U* was calculated for each dependent variable category. For the remaining categories (e.g., setting, diagnosis, and intervention type) the weighted Tau-*U* of each study or participant was a weighted mean calculated using the online calculator.

DHPS effect size. To estimate the magnitude of effects using a parametric effect size, we calculated a DHPS effect size (Hedges et al., 2012). DHPS uses a hierarchical model to produce a between-subjects effect size. This method can be used to calculate an effect size for studies that have at least three participants (i.e., cases; Marso & Shadish, 2014). Fourteen studies from our review were excluded from DHPS analysis because the designs did not meet these requirements (e.g., reversal with only one participant and multiple baseline across behaviors within one participant). Ten studies met the requirements, and the raw data were entered into Statistical Package for the Social Sciences (SPSS) version 26 using the macro developed by Marso and Shadish (2014). The macro computed the DHPS value for each study by dependent variable category as well as the variance of *g*, standard error, and degrees of freedom. Due to the small number of cases typically found in single-case studies, we constructed a symmetric confidence interval using a *t* distribution instead of the standard normal reference distribution. We used the following formula to calculate the confidence intervals, where $t(v)$ is the 97.5% critical value from a *t* distribution with *v* degrees of freedom (Pustejovsky et al., 2014):

$$g_{AB} \pm V_g \times t_{\frac{a}{2}v}.$$

STATA Software Version 16 was used to calculate a summary effect size and heterogeneity for individual DHPS values. Moderator analyses were run for each dependent variable category. Forest plots for both Tau-*U* and DHPS effects were created. In addition, using the DHPS values, a funnel plot for publication bias was created.

Inter-rater reliability. The first author trained two coders to evaluate 100% of the identified articles for each level of the exclusion and inclusion screening process. A third rater independently reviewed articles when there was a disagreement, and all raters came to a consensus. Prior to coding, the coders were trained on the coding manual with three articles at the beginning of the process that were not included in this study. Inter-rater reliability was calculated for percent agreement by dividing agreements by agreements plus disagreements and multiplying by 100. At the title/abstract and full-text review stage, inter-rater reliability between the two coders was 92% and 97% respectively. Variable coding inter-rater reliability was calculated for each category and then combined for an average score. This was completed for 100% of the articles by the first author and fourth author and yielded 97% agreement. For data analysis (effect size calculations), inter-rater reliability was 99% agreement for 22% of the graphs and effect size calculations.

Results

Database searches of PsychINFO via ProQuest Education Journal, PubMed, and ERIC through 2018 (and subsequent review using the procedures using the procedures listed above) along with quality of evidence evaluation resulted in the identification of 24 single-case design articles. The studies included a total of 65 participants with ASD who had both baseline and instructional phases completed and met the inclusion requirements. An overview of the included studies is provided in Table 1.

Descriptive Analysis

Participant characteristics. The 24 studies included between one and seven participants with ASD. The mean participant age in years was 15:7 (years:months), with a range from 13 to 21 years. Fifty-one participants were male (78%) and 14 were female (22%). Twenty-four participants were White (37%), 10 were African American (15.38%), five were Latinx (8%), one participant was multiracial (2%), and one participant was Asian (2%). Race/ethnicity data were not reported for 24 participants (36.9%). The majority of participants were described as having a diagnosis of ASD ($n = 34$; 51%) and six participants (13%) were described by authors as having a milder form of ASD (e.g., Asperger syndrome). Twenty-five participants (36%) were identified as having ASD and a comorbid diagnosis—for all 25, this was an ID. Fourteen participants (22%) were described as having complex communication needs or exhibiting severe limitations in speech; the remaining 51 participants (78%) were described as having the ability to use speech during interactions with others.

Intervention characteristics. Seven studies (29%) were coded as *peer-directed interventions*. Examples of peer-directed interventions include Carter et al. (2005), where peers

Table 1. Summary of Coded Variables and Study Ratings.

Study	IV category	Setting	Design quality	Strength of evidence
Bambara et al. (2016)	CA	Cafeteria	Meets with reservations	Strong evidence
Carter et al. (2005)	PD	General Ed.	Meets with reservations	Strong evidence
Carter et al. (2017)	PD	General Ed.	Meets standards	Moderate evidence
Davis et al. (2010)	AD	Spled	Meets standards	Strong evidence
Gardner et al. (2014)	CA	Multiple	Meets with reservations	Strong evidence
Gaylord-Ross et al. (1984)	CA	Common space	Meets with reservations	Strong evidence
Halle et al. (2016)	CA	General Ed.	Meets standards	Moderate evidence
Haring & Breen (1992)	CA	Multiple	Meets with reservations	No evidence
Hochman et al. (2015)	PD	Cafeteria	Meets standards	Strong evidence
Huber et al. (2018)	PD	General Ed.	Meets standards	Moderate evidence
Hughes et al. (2002)	PD	Multiple	Meets with reservations	Strong evidence
Hughes et al. (2011)	CA	Cafeteria	Meets with reservations	Strong evidence
Hughes, Harvey, et al. (2013)	PD	Spled	Meets with reservations	Strong evidence
Hughes, Bernstein, et al. (2013)	CA	Multiple	Meets with reservations	Strong evidence
Koegel et al. (2012)	AD	Cafeteria	Meets standards	Moderate evidence
Koegel et al. (2013)	AD	Cafeteria	Meets with reservations	Moderate evidence
Morrison et al. (2001)	CA	Common space	Meets with reservations	Strong evidence
Plavnick et al. (2015)	AD	Spled	Meets with reservations	Moderate evidence
Plavnick & Dueñas (2018)	AD	Spled	Meets with reservations	Moderate evidence
Reilly et al. (2014)	CA	General Ed.	Meets with reservations	Strong evidence
Scattone et al. (2006)	AD	Cafeteria	Meets with reservations	No evidence
Schaefer et al. (2018)	PD	General Ed.	Meets standards	Strong evidence
Sreckovic et al. (2017)	CA	Cafeteria	Meets standards	Moderate evidence
Stauch et al. (2018)	CA	Spled	Meets with reservations	Strong evidence

Note. AD = learner with ASD directed; CA = combined approach; General Ed. = general education classroom; IV = independent variable; PD = peer-directed; Spled = special education classroom.

received training in social skills strategies for 2 to 4 days that included ongoing monitoring and feedback on the implementation of the targeted strategies. Alternatively, in Hughes et al. (2002), peers were simply told, "I'd like you to hang out with ____ as a friend" (p. 392) when they were in proximity to a participant with ASD; no additional prompting, feedback, or instruction was provided throughout the intervention. Of the seven peer-directed interventions, three provided brief training in social supports; for these interventions, peers were given a relatively short training (e.g., one orientation meeting) and directed to interact with the learners with ASD. The remaining four studies in the peer-directed category provided organized instruction that included ongoing monitoring, feedback, and opportunities for modeling and practice for peers. For example, Hughes, Harvey, et al. (2013) was successful in increasing interactions for three participants with ASD by training peers to interact with the participants and to set goals for the number of interactions that the partner would engage in each day.

Six (25%) studies were coded as *learner with ASD-directed* interventions. These interventions included, for example, Davis et al. (2010) in which the interventionist taught three participants with ASD to ask "others-focused" questions through the use of cue cards and included opportunities for practice with examples and nonexamples. In two other studies, researchers created social clubs based on the interests of the learners with ASD, and included adult facilitation (i.e., assistance in recruiting club-members and instruction in activities) during club activities (Koegel et al., 2013; Koegel et al., 2012). Four studies provided direct teaching or instruction and used instructional techniques such as social narratives (Scattone et al., 2006), charts/cards (Davis et al., 2010), and video instruction (Plavnick et al., 2015). Koegel et al. (2012, 2013) found increases in both initiations and engagement by providing minimal assistance to the learners with ASD by facilitating clubs based on the participants' interests.

Finally, 11 (46%) studies were coded as *combined-approach* interventions. For example, Bambara et al. (2016) taught participants with ASD to perform skills listed on cue cards during interactions, and also provided training for the peer partners using role-plays. Hughes et al. (2011) taught both the participants with ASD and peer partners to use a communication book during social interactions. The communication book included a variety of topics and questions for conversation, and peers and participants were taught procedures for asking questions and providing responses.

Setting characteristics. The majority of the studies were conducted in the cafeteria ($n = 7$; 29%). Studies also were conducted in the general education classroom ($n = 6$; 25%), special education classroom ($n = 5$; 21%), multiple settings ($n = 4$; 16%), or common areas ($n = 2$; 8%).

Outcome measure characteristics. All studies reported at least one dependent variable for social interactions for the participant(s). A large proportion of the studies measured social interactions as any conversational act ($n = 14$), while some studies measured specific social interaction skills such as initiations ($n = 11$), asking follow-up questions ($n = 2$), responses ($n = 7$), and commenting ($n = 4$). Additional social interaction skills such as others-focused conversation ($n = 2$) and engagement ($n = 4$) were also measured. Fifteen of the 24 studies measured multiple dependent variables. Dependent variables were measured by rate ($n = 3$), partial interval recording ($n = 11$), frequency ($n = 9$), quality ($n = 3$), momentary time sampling ($n = 2$), and duration ($n = 3$).

Research design and characteristics. Nearly all of the studies ($n = 22$; 92%) were multiple-baseline across participants or across behaviors designs. The remaining two studies (8%) employed withdrawal/reversal designs. Social validity was reported in 58% of the studies ($n = 15$). Social validity was assessed across participants, peers, teachers, staff members, parents, and research facilitators. Multiple methods such as questionnaires, interviews, and surveys were used within the studies. Although the outcomes of the interventions were generally viewed positively by participants, a small number of learners with ASD rated the interventions unfavorably or provided statements of neutrality regarding the outcomes or necessity of instruction (e.g., Bambara et al., 2016; Carter et al., 2017; Davis et al., 2010).

Procedural integrity was reported in 63% ($n = 15$) of the studies. Sixty-three percent ($n = 10$) of those studies averaged above 90% for treatment fidelity with the remaining 38% ($n = 5$) averaging 80% to 90%. All 24 studies provided a report of IOA with 18 reporting an average of 90% agreement or higher, and six reporting an average of 80% to 90% agreement. See Supplemental Materials for additional details.

Strength of Evidence

Of the 24 studies that met standards, 13 demonstrated strong evidence, nine demonstrated moderate evidence, and two demonstrated no evidence according to the WWC criteria. The most common reason for studies falling in the "no evidence" category was insufficient demonstrations of basic effects at three or more different points in time (i.e., a functional relation in a multiple baseline design was not supported). For studies with multiple dependent variables, each variable was evaluated individually (see Table 1).

Table 2. Results of Studies by Participant, Intervention, and Intervention Outcomes Characteristics.

Measure	Number of cases[a]	Tau-*U*	Size of effect
Participant characteristics			
Diagnosis			
Autism	34	0.90	Medium to large
HFA	6	0.96	Medium to large
ASD and comorbid	25	0.93	Medium to large
Communication characteristics			
Speech	51	0.91	Medium to large
CCN	14	0.92	Large
Intervention characteristics			
Type			
Peer-directed	7	0.92	Medium to large
Learner with ASD-directed	6	0.96	Large
Combined-approach	11	0.90	Medium to large
Setting			
Special education	5	0.97	Large
General education	6	0.82	Medium to large
Common spaces	2	0.99	Large
Cafeteria	7	0.95	Large
Multiple settings	4	0.97	Large
Outcome characteristics			
Conversation acts	12	0.94	Large
Initiations/requests	7	0.91	Medium to large
Follow-up questions	1	0.96	Medium to large
Responses	5	0.85	Medium to large
Commenting	3	0.96	Large
Others-focused conversation	2	0.89	Medium to large
Engagement	3	0.95	Large

Note. [a]The number of cases does not always equal the number of participants included in the analysis due to the variability across studies in the number of dependent variables reported. ASD = autism spectrum disorder; CCN = complex communication needs; HFA = high functioning autism.

Tau-U effect sizes. Tau-*U* was calculated for all 24 studies that met WWC design standards (with and without reservations) regardless of the level of evidence, as described in section "Method". The mean Tau-*U* value across studies was .92 (range = .65–1.0) indicating medium to large effects (Vannest et al., 2016). Table 2 displays the mean effect size by participant, intervention, and outcome characteristics (i.e., diagnoses and specific dependent variables measured within the study). When averaged across studies and participants, medium to large treatment effects were found for participants with three different diagnoses, for three types of interventions, and for seven different outcome variables.

Table 3. Effects for Dependent Variables and Heterogeneity of Effect Sizes.

Outcome variable categories	*n*	*M* (*g*)	95% CI	*Q*	*df*	*p*	I^2
Communication acts	6	4.249	[–0.453, 8.952]	31.18	5	<.001	99.72
Initiations/requests	6	2.028	[0.811, 3.245]	86.45	5	<.001	98.49
Follow-up questions	1	1.037	[0.497, 1.576]	–	–	–	–
Responses	3	2.192	[0.683, 3.702]	107.09	2	<.001	44.62
Others-focused conversation	1	1.651	[–1.393, 4.696]	–	–	–	–
Engagement	2	2.310	[1.507, 3.112]	4.82	1	.028	79.25

Note. n = number of studies; M = mean effect size.

DHPS effect sizes. Study level effect sizes across the 10 studies that qualified for DHPS analysis ranged from 1.04 to 21.40 with a mean adjusted effect size of $d = 2.07$; 95% confidence interval (CI) [1.57, 2.57]. This mean effect size is large according to commonly used benchmarks (Cohen, 1988). Mean effect sizes across outcome variable categories and the results from the heterogeneity test to examine the variation in effect sizes among studies are presented in Table 3. The mean effect sizes are 4.25 for communication acts, 2.03 for initiations and requests, 1.04 for follow-up questions, 2.19 for responses, 1.65 for others-focused conversation, and 2.31 for engagement—indicating large effects for all the variables based on Cohen's criteria (Cohen, 1988). The Q values for all outcome variable categories were significant across the studies and approximately 44.92% to 99.72% of the variability in effect sizes was due to heterogeneity rather than sampling errors (Borenstein et al., 2009). Figure 1 shows the forest plot of Hedges' g for the dependent variables.

Figure 2 displays two forest plots for the dependent variable categories. The top forest plot shows the Tau-U results, and the bottom forest plot shows the DHPS results. The bottom forest plot displays the DHPS effect sizes for each study by dependent variable. The confidence intervals are not visible for the majority of studies due to an outlier (Hughes, Bernstein et al., 2013).

Figure 3 shows the funnel plot for the DHPS effect sizes. The outer lines indicate the region within which 95% of study effects should lie assuming no bias. The plot shows a symmetrical distribution of effect sizes around the mean effect size.

Discussion

The primary purpose of the current meta-analysis was to assess the effects of social skills instruction for adolescents with ASD in public school settings. More specifically, we sought to (a) identify what social skill interventions have been studied for adolescents with ASD, (b) determine what social skills have been targeted, (c) evaluate the quality of the designs and strength of evidence, and (d) determine the effects of studies as measured by two effect size estimates. It should be noted that only 63% of studies reported on treatment fidelity results.

Given the large number of studies that did not report treatment fidelity, it is unclear whether the procedures were implemented as described, and the results should be interpreted with caution.

Characteristics of Interventions

Each type of intervention (i.e., peer-directed, learner with ASD-directed, and combined-approach) was effective in increasing social interactions for participants with ASD. Previous research suggests that an advantage of peer-mediated interventions is the minimal time needed to train peers to implement the intervention (Carter et al., 2011). For the studies in this review, social interactions among participants increased despite the minimal training and direction provided to the participating peers. Unfortunately, when solely relying on peers to implement the intervention, participating peers may lose interest and not perform the skills consistently over time. To address this challenge, Carter et al. (2005) recommended targeting groups of peers for training instead of a single peer buddy. In the interventions directed at learners with ASD, each of the participants demonstrated increases after the implementation of the intervention.

Combined-approach interventions (i.e., interventions that provided training for both the peer partner and the student with ASD) were used in a majority of the studies across a variety of settings (e.g., cafeteria, common space, and classroom). After instruction is provided, combined-approach interventions allow adults to take on a less intrusive role during the intervention, with the peer partners taking on new communication roles. Using elements of specific social skill instruction for the learner with ASD, combined with the presence of peers and adults in supportive roles, students may be more likely to acquire and generalize skills to other settings and persons (Carter et al., 2014). It is interesting to note that though each study used peers and direct instruction/adult facilitation, the direct instruction strategy varied. For example, studies used instructional techniques such as explicit instruction (Reilly et al., 2014),

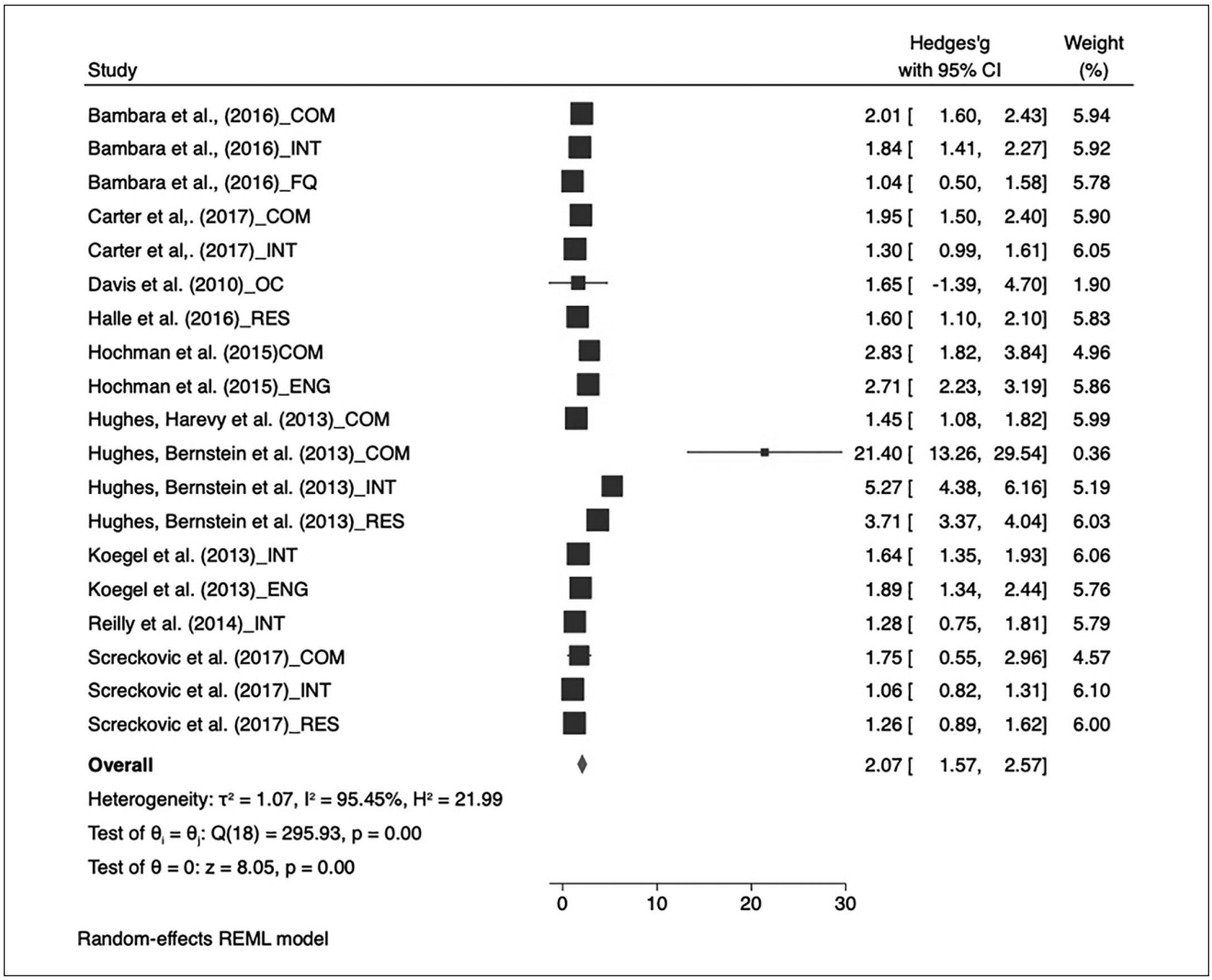


Figure 1. Forest plot of Hedges' *g* for dependent variables.

cue cards/communication books (Bambara et al., 2016; Hughes et al., 2011), and video instruction (Halle et al., 2016). The variation in the social skill instruction among the studies suggests that a variety of approaches can be used as long as they provide adequate support for the learning of the target skills.

Outcome Characteristics

The majority of studies measured frequency of initiations and/or responses made by the participant with ASD. Previous research has found that initiations are an especially challenging skillset for individuals with ASD (Paul et al., 2009). A promising finding in this review was that interventions focused on increasing initiations showed positive results. Only 13 studies measured high-level social skills such as follow-up questions, others-focused conversation, engagement, and commenting. Despite this, the current analysis revealed that in studies in which these skills were measured, each yielded medium–large Tau-*U* and large DHPS mean values. At present, research has focused on the most easily measured aspects of interactions (e.g., frequency of initiations and responses; Walton & Ingersoll, 2013). There has been limited research on more nuanced elements of social interaction, such as partner-focused questions and others-oriented comments that are critical to successful interaction during adolescence and adulthood (Smith, 2005). In Stauch et al. (2018), higher order skills such as joining or extending a conversation were targeted. The participants, however, had previously participated in a 4-month intervention study (Plavnick & Dueñas, 2018), and therefore were deemed to have the necessary prerequisite skills (e.g., asking social questions, commenting, and complimenting peers).

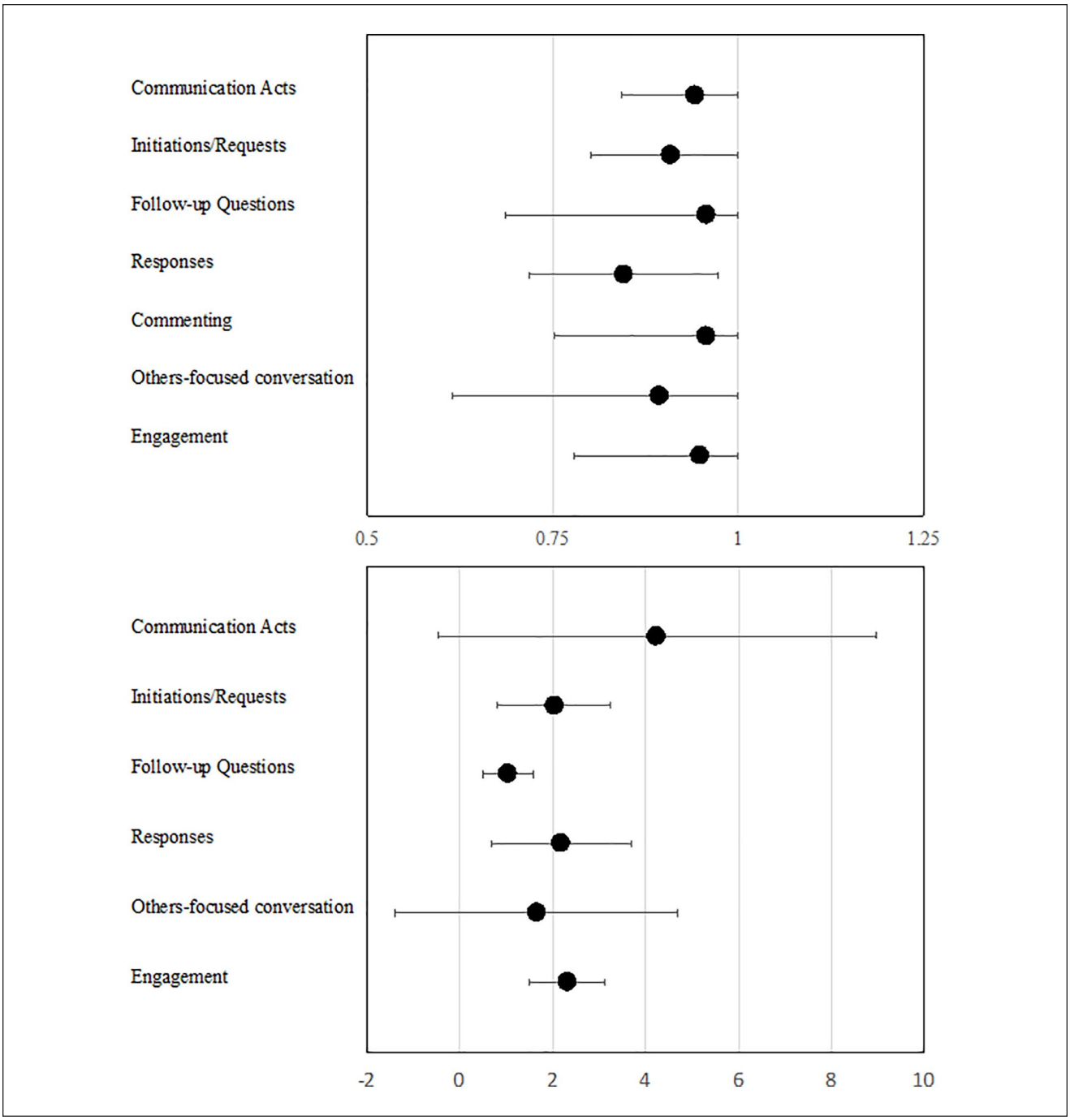


Figure 2. Top figure shows a forest plot of Tau-*U* data for dependent variables and the bottom figure shows a forest plot of DHPS effect sizes for dependent variables.

A number of important social skills have received limited attention to date including showing empathy, conversing about other's interests, compromising with others, and moving from one topic to another (*Vineland Adaptive Behavior Scales–Third Edition*; Sparrow et al., 2016). The studies in our review focused primarily on more basic discrete skills such as initiating conversation or responding during an interaction. Indeed, if authentic interactions between adolescents with ASD and their peers is the goal, interventions must target higher level social skills (Reilly et al., 2014). Currently, there is only limited information to develop interventions that would address the sophisticated communication behaviors observed in typical adolescent conversation.

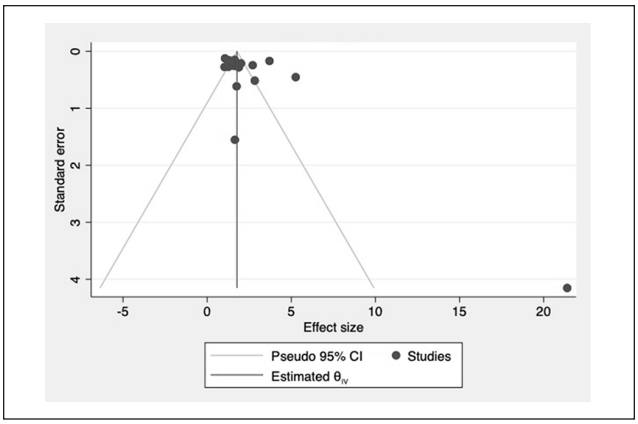


Figure 3. Funnel plot for Hedges' *g* data.

Study Quality and Evidence

Overall the strength of the evidence is strong per visual analysis, and this is supplemented with the results for both Tau-*U* and DHPS effect sizes. There may be moderating effects by outcome variable. Specifically, the outcome variable of communication acts had the largest standardized effect size (DHPS); however, given the large confidence interval and likelihood of selective reporting, this finding should be taken with caution. Publication bias may be at play as studies with negative findings are likely to be rejected (Tincani & Travers, 2018). Thus, these effects need to be interpreted with caution. We cannot say for certain that these interventions will be effective for all learners with ASD.

Our sample included an outlier (Hughes, Bernstein, et al., 2013), which had a large confidence interval. Indeed, all effect sizes are likely inflated. First, all Tau-*U* values are medium to large according to reported standards (Vannest & Ninci, 2015). This could be due to a limitation of Tau-*U*, in that it only measures overlap (i.e., not magnitude). Yet, when comparing Tau-*U* to the DHPS effect sizes, which do capture magnitude (i.e., standardized *d*-estimate), the results are similar. There were some differences as DHPS does not cap at 1.00, whereas, Tau-*U* does (i.e., an overlap index), which means that these two effect size estimates are not completely comparable.

Limitations, Future Research, and Implications

There are some limitations worth noting. Although the studies in this review were evaluated for whether or not procedural fidelity was reported, we included studies in our analysis that did not report procedural fidelity data. Future reviews should consider procedural fidelity as a required design standard. Without fidelity data, it is unclear whether procedures were implemented as intended and with replicable precision. Therefore, the internal validity of a study is compromised as the specific environmental variables hypothesized to change behavior are undocumented. Furthermore, in this review, many of the studies utilized package interventions. A task analysis assessing treatment fidelity could allow other researchers to replicate procedures more easily and provide insight regarding the active ingredients of the intervention.

Second, we employed a nontraditional approach to some of the datasets by combining similar phases that were adjacent (e.g., BC phases of a sequential intervention package), which could be viewed as a limitation. This approach, however, addressed our fourth research question in the most parsimonious way. We were interested in determining the effects of interventions and packages on social skills. Examining the effects of each intervention component was beyond the scope of the current analysis, yet could be an area for future research. Previous work has set precedent for comparing noncontiguous phases in single-case meta-analysis (Neely et al., 2016).

Third, we selected two approaches to evaluate the efficacy of the studies, and each approach has its limitations. Although Tau-*U* is regarded as a robust measure for single-case, it is a nonparametric measure of nonoverlapping data (Vannest & Ninci, 2015). Therefore, it does not capture the magnitude, level change, immediacy of effect, account for variability, or take into account the vertical analysis needed to assess experimental control in a multiple baseline design. Furthermore, though DHPS is recommended as a parametric effect size measure for single-case-design, this method does not analyze trend. In addition, we could only include 10 studies in the DHPS analysis due to studies needing to have at least three cases, which ruled out multiple baseline across behaviors/skills designs (e.g., Hughes et al., 2011). A final limitation of the DHPS effect size is its application to multiple-probe designs. Specifically, multiple-probe designs have a limited number of baseline data points, which result in less degrees of freedom. The small number of data points provided in some of the studies (e.g., Davis et al., 2010), may be impacted by time or history effects, leading us to be less confident of those results. In addition, unbalanced numbers of data points between baseline and intervention phases (e.g., Hughes, Bernstein, et al., 2013) may result in larger, and inflated, effect sizes. Multiple-probe designs—even when meeting minimum WWC requirements—may not fully meet the assumptions of a hierarchical statistical model. Although there is limited precedent for interpreting DHPS effect sizes for single-case designs, our results are in line with previous meta-analyses that have applied these methods (e.g., Brock et al., 2017).

Although the findings of the current meta-analysis provide additional evidence that social skill interventions have a promising evidence-base for adolescents with ASD, with regard to participant demographics, race and ethnicity was largely underreported in the studies. For more than a third

of the participants, race and ethnicity data were not reported. Furthermore, most of the participants were White. Due to the small number of studies, the relatively homogeneous nature of the participants, and underreporting of demographic information (e.g., race of participants), little can be determined in regards to what types of interventions were most effective for what students. Future research should report on variables that affect social interactions including race and ethnicity, as these have implications for cultural norms and expectations. Social skills are innately culturally specific, and simply reporting race and ethnicity is a logical and necessary first step. However, future research should not stop there. With an eye toward culturally adapting social skill interventions, future intervention research should purposefully include more diverse samples including race, ethnicity, socioeconomic status, and other contributing factors (Safer-Lichtenstein et al., 2019). Further investigation is also needed to determine which social outcome measures are the most important for this age group, as key social skills for older students may be different from those targeted for younger students (Walton & Ingersoll, 2013). Social interactions are complicated at any age, and adolescence may pose special challenges that require sophisticated skills (Smith, 2005). Establishing what social skills will increase positive social interactions may contribute to an enhanced quality of life for these individuals. For example, some adolescents may greatly benefit from interventions focused on more pragmatic social language skills as opposed to initiating and responding.

Additional research is needed to consider the variability among participants and need for modifications in the intervention. For example, despite learner with ASD-directed interventions having the flexibility to be individualized to the participant, no modifications were made to these interventions for students who may have needed additional supports. Similarly, adaptations may be needed to ensure social skill interventions are accessible for students with limited speech. For example, participants with complex communication needs may benefit from augmentative and alternative communication (AAC) within the intervention, particularly to support a wider range of communication acts (e.g., initiating topics and requesting information; Carter et al., 2014; Smith, 2015). Finally, a comprehensive view of social validity includes consideration of the acceptability of the treatment to the participants, the outcomes achieved, and the perceived value of the intervention. As a small number of learners with ASD viewed the interventions unfavorably or were neutral regarding the intervention, future consideration should be given to the interests of the learner and aim to incorporate motivating activities into the social interaction.

The current meta-analysis extends the literature and past reviews by (a) evaluating single-case studies conducted in public-school settings and (b) including adolescents with a range of ASD diagnoses and communication characteristics. The meta-analysis provides evidence that social interaction interventions can have a positive impact for adolescents with ASD. Although there has been an increase in studies addressing social skill interventions for individuals with ASD, continued research is urgently needed for adolescents with ASD at the middle and high school levels (Chung et al., 2012; Smith, 2005), across various races and ethnicities, and for individuals with limited speech who would benefit from AAC supports. Future research must support education professionals in implementing effective social skill interaction interventions for adolescents with ASD in real-world, diverse school settings.

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Supplemental Material

Supplemental material for this article is available online.

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